TELECOMMUNICATIONS NETWORK OPERATIONS, CUSTOMER EXPERIENCE & SERVICE ASSURANCE ANALYTICS PLATFORM

Capstone Project Report

Snowflake | Informatica IDMC | Snowpark Python | Streamlit | Cortex Analyst

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1. Execution Summary

The project delivers a governed telecom analytics platform that unifies seven operational datasets across customer, subscription, usage, network, incident, ticket and event domains. IDMC performs upstream profiling and cleansing, Snowflake implements the medallion warehouse, Snowpark calculates explainable rule-based risk, and Streamlit presents operational and analytical views.

Outcome	Verified implementation
Scale	6,900 source rows across 7 datasets; 6,886 rows accepted after IDMC routing.
Warehouse	4 dimensions, 4 facts, 4 core marts, 10 KPI views and 6 Cortex semantic views.
Incremental flow	10-row Usage file with 8 new IDs and 2 changed IDs processed through Stream, procedure, SCD Type 2 and Gold MERGE.
Operational analytics	Main Streamlit dashboard, explainable risk view and a second telecom CRM dashboard.
Security	Three business roles, customer-data masking and secure reporting views.
Project delivery	Five-day sprint with 13 of 13 planned tasks completed by Day 5.

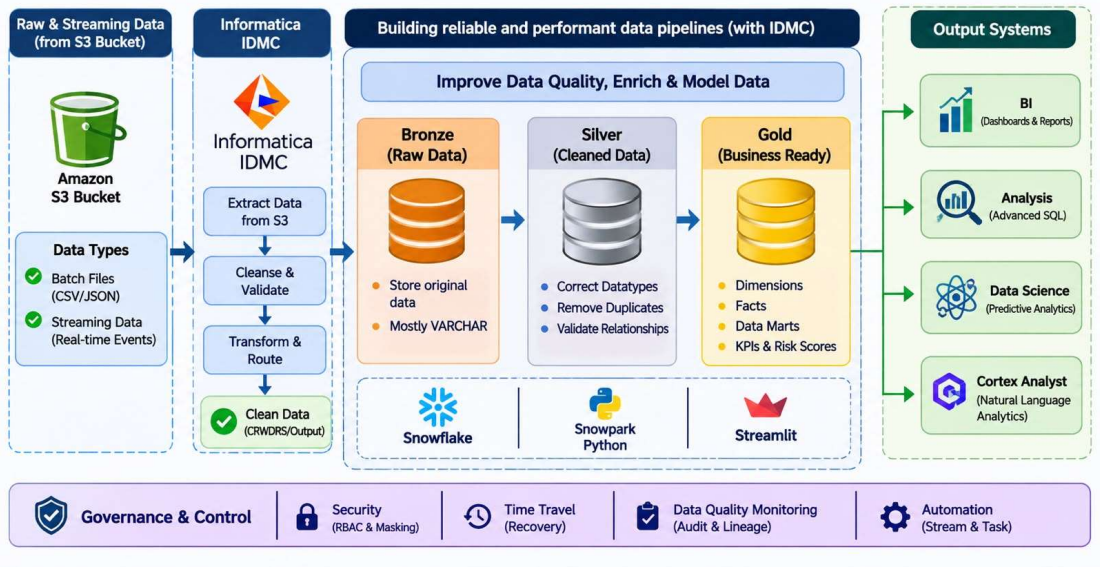


Fig. 1.1: Implemented platform architecture from source delivery to governed consumption

2. Problem Statement

Telecom operations data was distributed across seven source files, preventing teams from tracing a service problem from a network event through incident impact, ticket handling and customer usage. The data also included controlled nulls, duplicates, impossible timestamp sequences, outliers and mixed date formats. Without a governed model, KPI definitions, operational triage and customer-impact analysis were inconsistent.

- No single analytical path connected network elements, events, incidents, tickets, subscriptions and customers.
- Raw records could not be trusted without explicit profiling, exception routing and count reconciliation.

- Operational staff needed proactive alerting and investigation views, while business users needed curated, masked consumption views.
- The solution had to preserve incremental Usage history without double-counting historical versions.

3. Project Objectives

The objectives combine the trainer guide requirements with the implementation evidenced in the final presentation.

Category	Objectives
Data integration	Ingest six CSV sources and one JSON event source; preserve source metadata and reconcile counts.
Data quality	Profile, standardise, deduplicate, validate business sequences and route invalid records.
Warehouse	Implement Bronze, Silver and Gold layers with conformed dimensions, facts and marts.
Incremental processing	Process a second Usage file through COPY INTO, Stream, SCD Type 2 and Gold MERGE.
Analytics	Measure usage, incident duration, customer impact, MTTR, ticket resolution, SLA and alarms.
AI and risk	Provide six Cortex semantic views and an explainable Snowpark network-risk score.
Security	Apply RBAC, masking policies and secure views.
Presentation	Deliver Streamlit dashboards for KPI, risk, exception and alert-triage use cases.

4. Business Understanding

The platform supports multiple telecom operating perspectives while retaining common definitions and shared conformed dimensions.

Team	Meaning in the Project
Network Engineering	Investigates raw and conformed events, incidents, latency, packet loss, CPU and traffic metrics.
Service Assurance	Monitors reliability, open alarms, incident impact, ticket queues and SLA compliance.
Customer Operations	Uses customer, subscription and ticket context to understand service impact.
Telecom Management	Consumes curated KPI and risk views without direct raw-table access.
Data Engineering	Maintains ingestion, data quality, medallion transformations, incremental loads and recovery.
Analytics Users	Use dashboards and Cortex Analyst to ask governed business questions without writing SQL.

5. Source Data & Data Understanding

The source package contains seven datasets and 6,900 records. Each dataset was assigned an explicit business grain and downstream purpose before transformation.

Source	Business Purpose	Grain	Key Columns
customers.csv	Customer identity and segmentation	One row per customer	customer_id, region, segment, contact fields
subscriptions.csv	Service ownership and status	One row per subscription	subscription_id, customer_id, service_type, dates
network_elements.csv	Physical/logical network inventory	One row per network element	network_element_id, parent_element_id, type, region
usage_transactions.csv	Voice, SMS, data and charge activity	One row per usage transaction	usage_transaction_id, subscription_id, usage_ts, measures
trouble_tickets.csv	Customer-reported service work	One row per ticket	ticket_id, network_incident_id, priority, SLA
network_incidents.csv	Service-impacting incidents	One row per incident	network_incident_id, primary_element_id, severity, timestamps
network_events.json	NMS alarms and telemetry context	One row per event	event_id, network_element_id, severity, metric_context

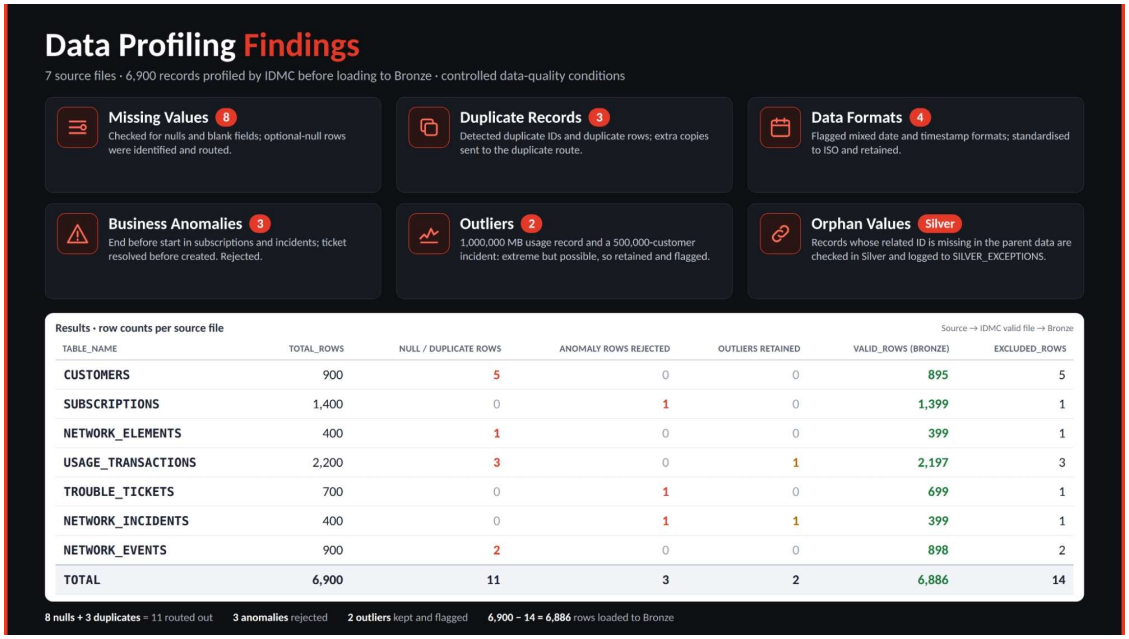


Fig. 5.1: IDMC profiling results and source-to-Bronze row reconciliation

6. End-to-End Architecture

The implemented journey is S3 to IDMC, named Snowflake stage, Bronze, Silver, Gold, Consumption and Streamlit. A separate incremental Usage path appends a second file to Bronze and consumes only new rows through a Stream. CONTROL holds procedures, tasks, exceptions and masking policies, while SEMANTIC holds the Cortex Analyst semantic views.

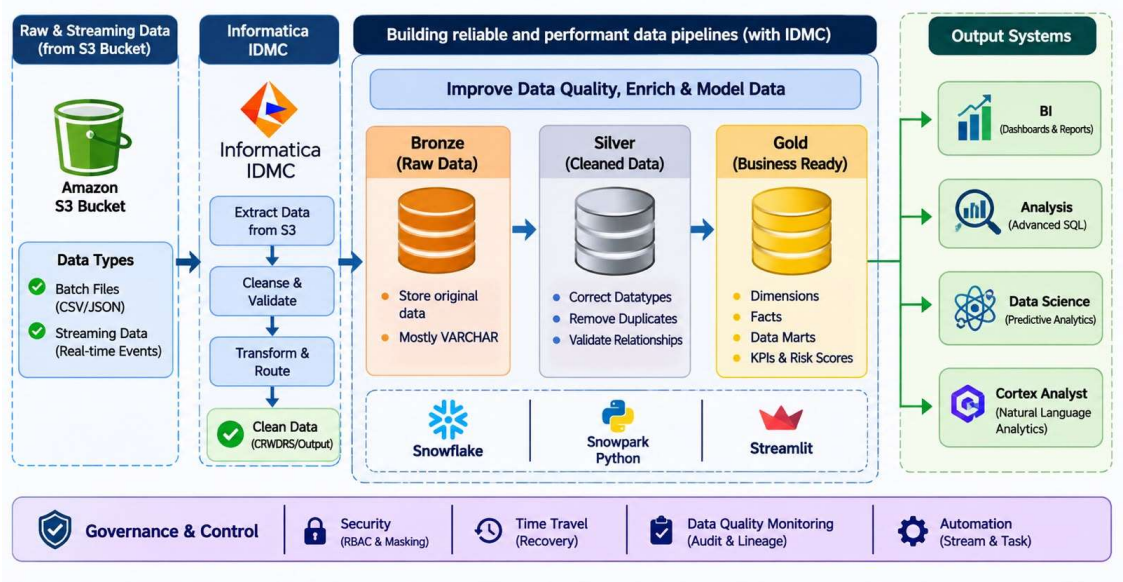


Fig. 6.1: End-to-end architecture and governance controls



Fig. 6.2: Validated Silver records transformed into Gold marts, KPIs and risk scores

7. IDMC ETL & Data Quality Layer

IDMC is the authoritative upstream profiling and cleansing layer. Separate mappings were created for the seven datasets, with clean outputs routed for Snowflake loading and invalid or duplicate rows retained as quality evidence.

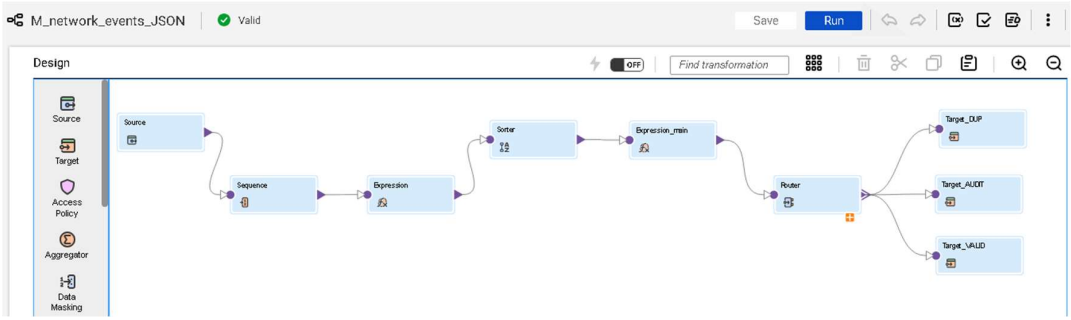


Fig. 7.1: IDMC mapping for JSON network events

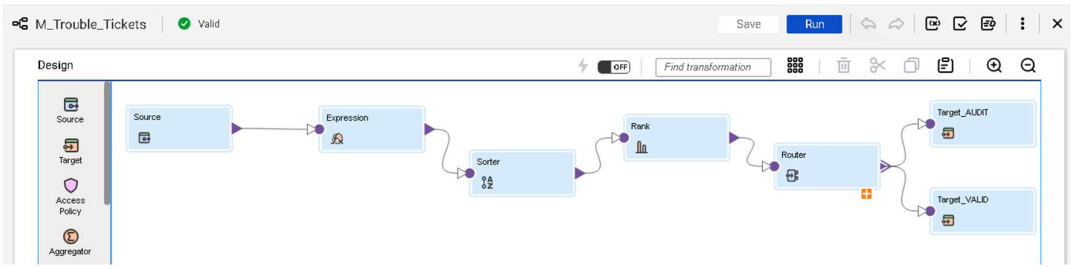


Fig. 7.2: IDMC mapping for trouble-ticket validation

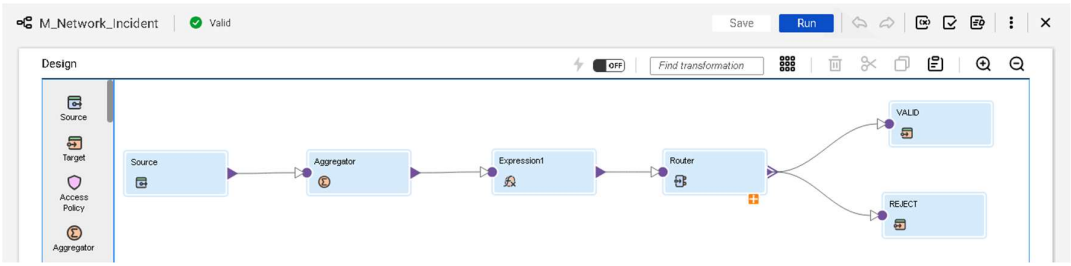


Fig. 7.3: IDMC mapping for network-incident validation

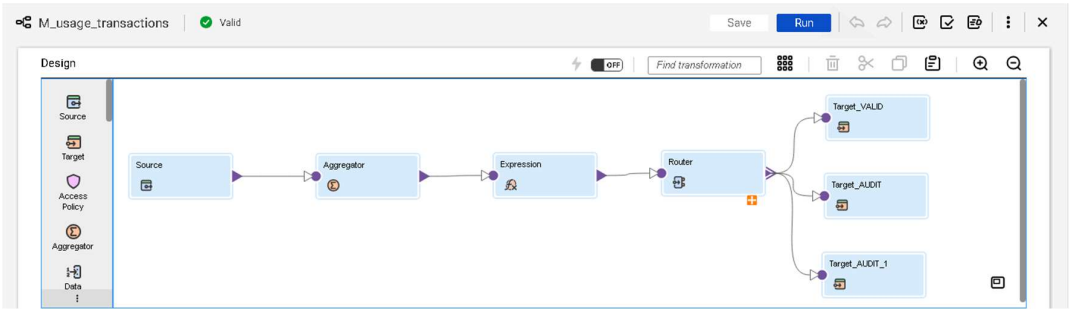


Fig. 7.4: IDMC mapping for usage-transaction validation

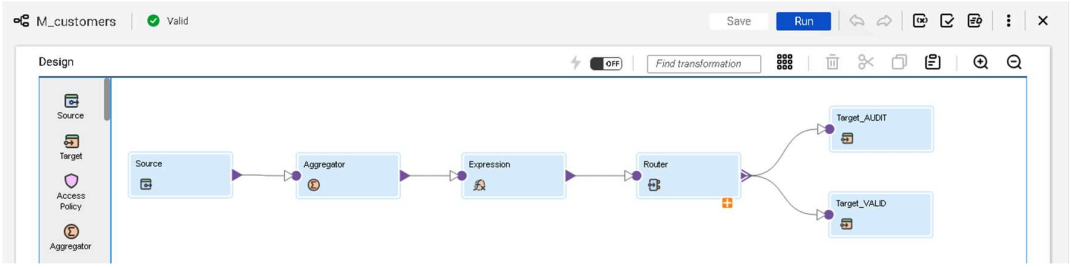


Fig. 7.5: IDMC customer cleansing and valid-record route

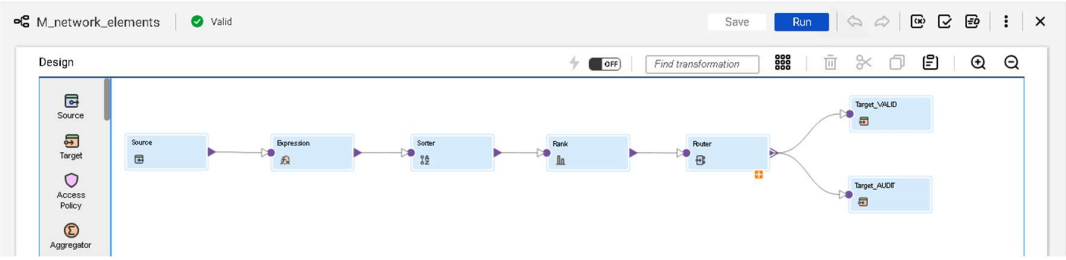


Fig. 7.6: IDMC network-element cleansing and valid-record route

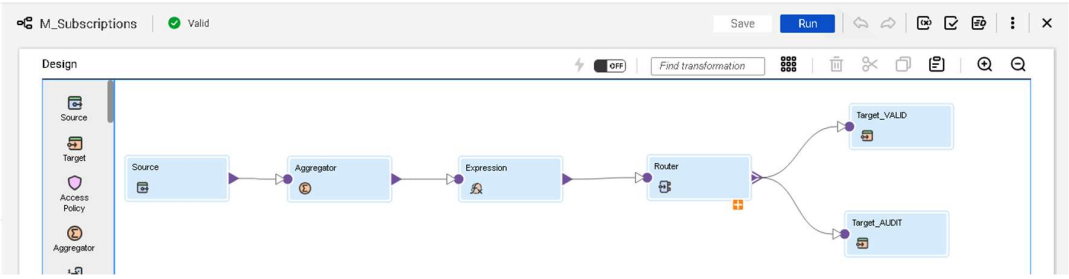


Fig. 7.7: IDMC subscription cleansing and valid-record route

Instance Name	Location	Subtasks	Start Time	End Time	Rows Processed	Status
m_network_in...	capstoneproject		Nov 2, 2026, 7:56 ...	Nov 2, 2026, 7:56 ...	400	Success
M_network_ev...	capstoneproject		Nov 2, 2026, 5:13 ...	Nov 2, 2026, 5:14 ...	900	Success
M_customers-1	capstoneproject		Nov 2, 2026, 5:00 ...	Nov 2, 2026, 5:00 ...	899	Success
M_network_el...	capstoneproject		Nov 2, 2026, 4:55 ...	Nov 2, 2026, 4:55 ...	400	Success
usage_transa...	capstoneproject		Nov 2, 2026, 4:44 ...	Nov 2, 2026, 4:44 ...	2199	Success
M_Subscriptio...	capstoneproject		Nov 2, 2026, 4:36 ...	Nov 2, 2026, 4:37 ...	1400	Success
M_Trouble_Tic...	capstoneproject		Nov 2, 2026, 4:26 ...	Nov 2, 2026, 4:27 ...	700	Success

Fig. 7.8: Successful IDMC mapping executions and processed-row evidence

8. Data Cleansing & Transformation Logic

Cleansing rules were designed around correctness, explainability and controlled routing. Correctable formats were standardised and retained; duplicates and impossible sequences were excluded; extreme but possible values were retained with an outlier flag. Silver then performed datatype conformance, relationship checks and JSON extraction.

Condition	Handling	Reason
Mandatory key is null	Reject or exception route	A record without its business key cannot be reconciled.
Duplicate business key or duplicate row	Duplicate route	Prevents unintended double counting.
Date or timestamp format variation	Standardise	Preserves valid business meaning.
End before start / resolve before create	Reject	Business sequence is impossible.
Extreme but possible measure	Retain and flag	Keeps legitimate rare events visible.
Missing parent or foreign key	Silver exception	Protects dimensional and hierarchy integrity.

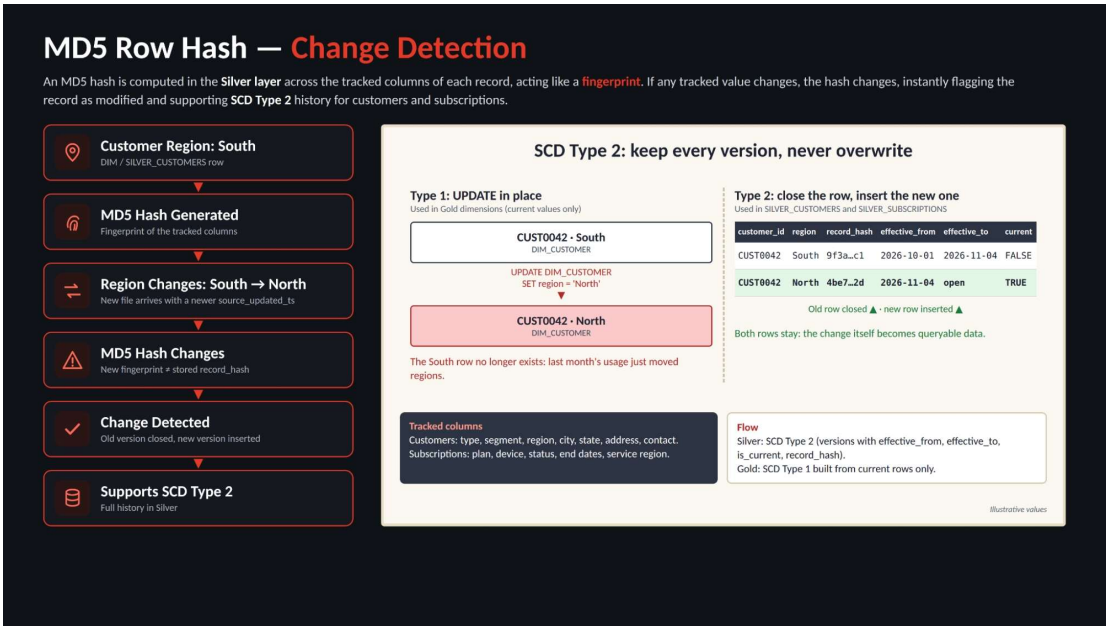


Fig. 8.1: MD5 row-hash change detection and historisation concept

9. Snowflake Architecture & Implementation

TELECOM_DB separates responsibilities into BRONZE, SILVER, GOLD, SEMANTIC, CONSUMPTION and CONTROL schemas. Raw files are loaded with source metadata. Silver tables apply typed business structures and preserve the Usage history. Gold tables use surrogate keys and conformed dimensions. Consumption exposes reporting-safe objects, while Control isolates operational logic.

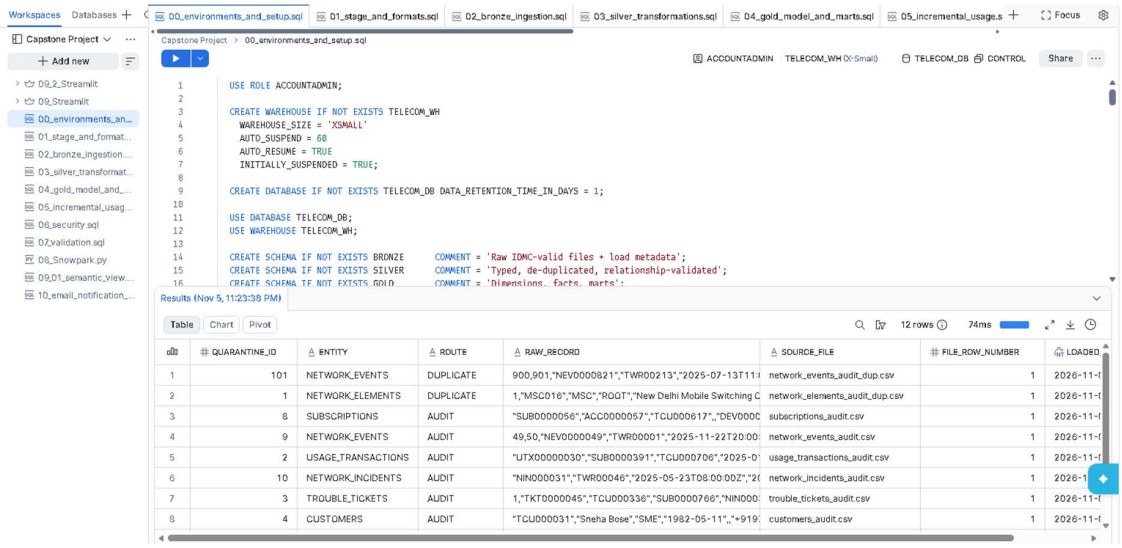


Fig. 9.1: Snowflake project scripts and schema-oriented implementation workspace

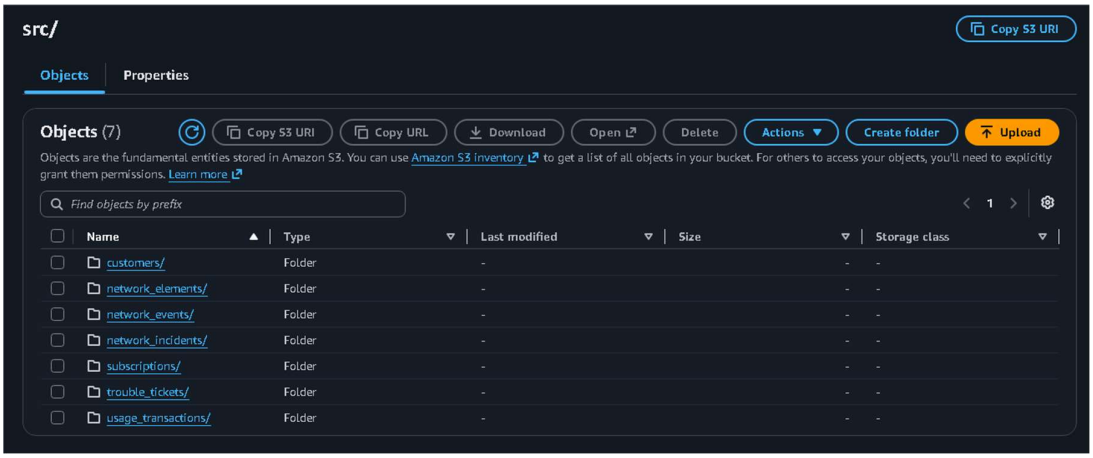


Fig. 9.2: Snowflake object and script management evidence

10. Medallion Architecture

The medallion pattern provides progressive quality and usability. Bronze stays source-aligned for traceability. Silver contains typed, deduplicated and relationship-validated records. Gold applies dimensional modelling and business calculations. The project deliberately keeps IDMC as the primary cleansing layer to avoid duplicating business-quality logic inside Snowflake.

Layer	Purpose	Representative objects
Bronze	Raw landing plus SOURCE_FILE and LOADED_AT	BRONZE_CUSTOMERS, BRONZE_USAGE_TRANSACTIONS, BRONZE_NETWORK_EVENTS
Silver	Typed and conformed records, JSON extraction and exceptions	SILVER_CUSTOMERS, SILVER_USAGE_TRANSACTIONS, SILVER_EXCEPTIONS
Gold	Dimensions, facts, marts and KPIs	DIM_CUSTOMER, FACT_USAGE, MART_NETWORK_RELIABILITY
Semantic	Business-friendly model for Cortex Analyst	Six TELECOM_*_SV semantic views
Consumption	Secure reporting boundary and persisted Snowpark output	V_USAGE_KPI, V_TICKET_SLA, NETWORK_RISK_SNOWPARK

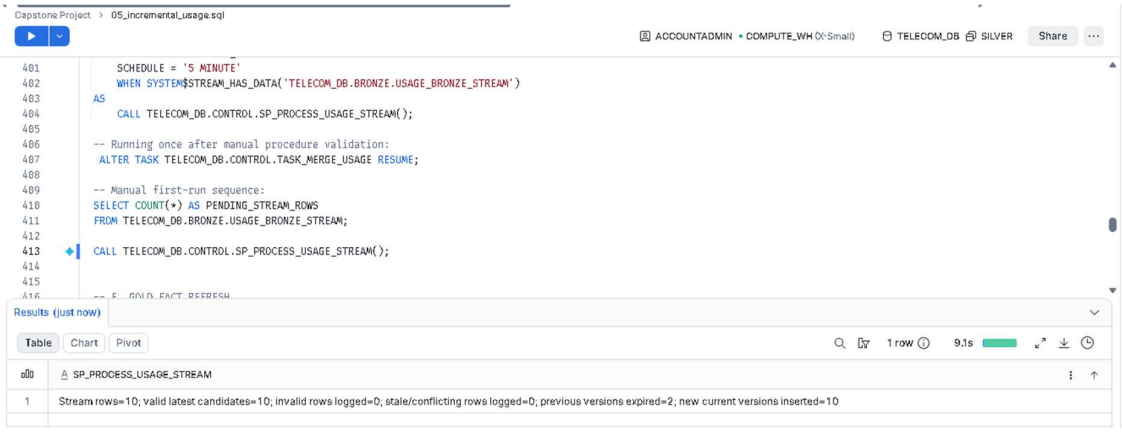


Fig. 10.1: Telecom warehouse flow from Silver through transformation tasks to Gold

11. Data Warehouse / Dimension Modelling

The implemented warehouse is a galaxy schema: four fact tables share four conformed dimensions. This structure supports cross-process analysis while preserving the natural grain of usage, incidents, tickets and events. DIM_DATE is role-playing for usage, incident start/end and ticket creation/resolution dates.

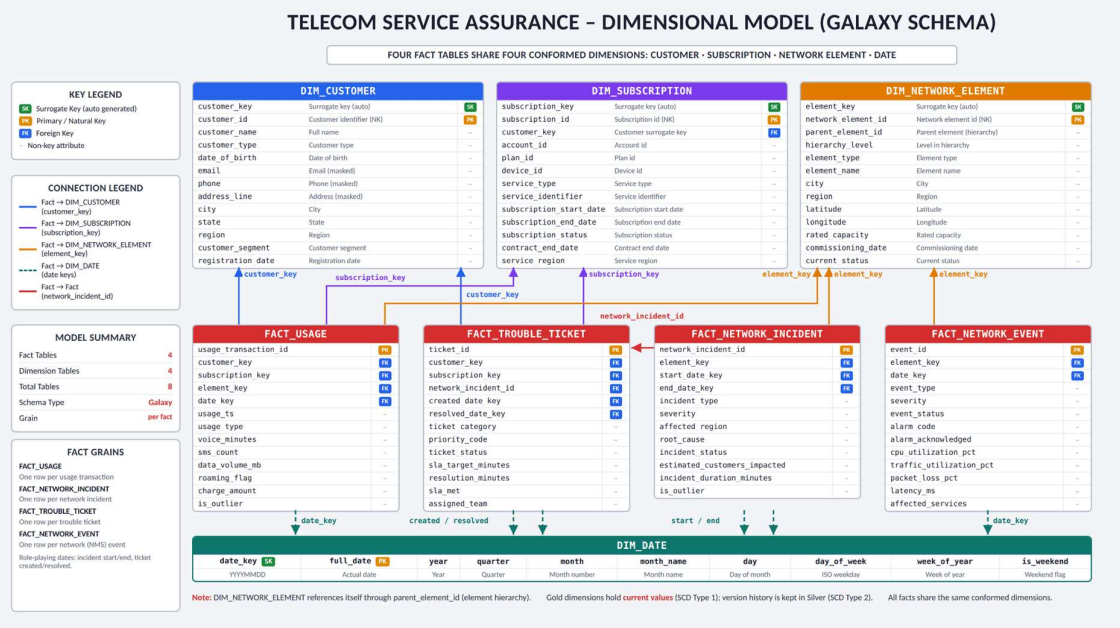


Fig. 11.1: Galaxy schema with four facts and four conformed dimensions

Fact	Declared grain	Primary analytical use
FACT_USAGE	One row per current usage transaction	Voice, SMS, data, roaming and charges
FACT_NETWORK_INCIDENT	One row per network incident	Frequency, severity, duration and impact
FACT_TROUBLE_TICKET	One row per trouble ticket	Resolution time, status and SLA
FACT_NETWORK_EVENT	One row per NMS event	Alarm, utilisation, packet loss and latency

12. Important SQL & Transformation Logic

The implementation uses Snowflake-native patterns that are safe to rerun and easy to validate: TRY_TO_* conversions, ROW_NUMBER-based deduplication, HASH/MD5 fingerprints, MERGE, Streams, Tasks, secure views and transactional stored procedures.

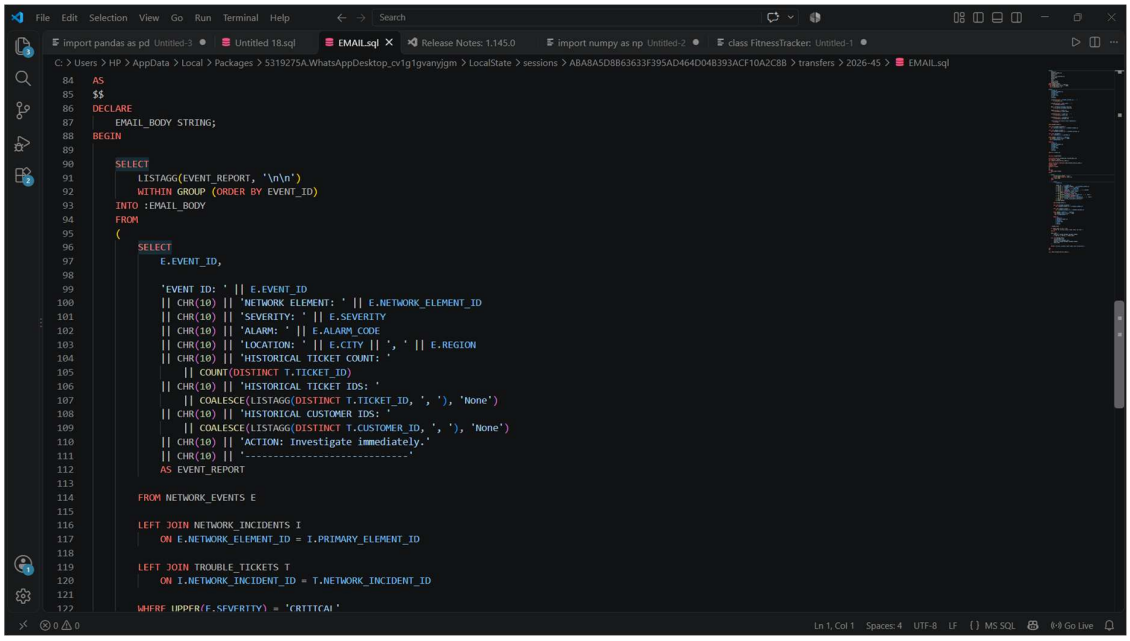
```
TRY_TO_TIMESTAMP(raw_usage_ts) AS usage_ts

ROW_NUMBER() OVER (PARTITION BY business_key ORDER BY version_ts DESC) AS rn

SYSTEM$STREAM_HAS_DATA('TELECOM_DB.BRONZE.USAGE_BRONZE_STREAM')

MERGE INTO GOLD.FACT_USAGE USING SILVER.VW_USAGE_TRANSACTIONS_CURRENT
```

The report intentionally provides representative patterns rather than inventing complete scripts. Exact SQL should be taken from the project repository and executed against the target account. The critical alert procedure additionally uses LISTAGG to assemble one context-rich report block per qualifying event.



```
AS
$$
DECLARE
    EMAIL_BODY STRING;
BEGIN

    SELECT
        LISTAGG(EVENT_REPORT, '\n\n')
        WITHIN GROUP (ORDER BY EVENT_ID)
    INTO :EMAIL_BODY
    FROM
    (
        SELECT
            E.EVENT_ID,

            'EVENT ID: ' || E.EVENT_ID
            || CHR(10) || 'NETWORK ELEMENT: ' || E.NETWORK_ELEMENT_ID
            || CHR(10) || 'SEVERITY: ' || E.SEVERITY
            || CHR(10) || 'ALARM: ' || E.ALARM_CODE
            || CHR(10) || 'LOCATION: ' || E.CITY || ', ' || E.REGION
            || CHR(10) || 'HISTORICAL TICKET COUNT: '
               || COUNT(DISTINCT T.TICKET_ID)
            || CHR(10) || 'HISTORICAL TICKET IDS: '
               || COALESCE(LISTAGG(DISTINCT T.TICKET_ID, ', '), 'None')
            || CHR(10) || 'HISTORICAL CUSTOMER IDS: '
               || COALESCE(LISTAGG(DISTINCT T.CUSTOMER_ID, ', '), 'None')
            || CHR(10) || 'ACTION: Investigate immediately.'
            || CHR(10) || '----------------------------'
            AS EVENT_REPORT

        FROM NETWORK_EVENTS E

        LEFT JOIN NETWORK_INCIDENTS I
            ON E.NETWORK_ELEMENT_ID = I.PRIMARY_ELEMENT_ID

        LEFT JOIN TROUBLE_TICKETS T
            ON I.NETWORK_INCIDENT_ID = T.NETWORK_INCIDENT_ID

        WHERE UPPER(E.SEVERITY) = 'CRITICAL'
```

Fig. 12.1: SQL logic that composes enriched critical-event email content

13. Telecom Usage, Incident, Ticket Delay & Network Reliability Analysis

The analytical layer exposes a connected view of usage, incident burden, event telemetry and ticket resolution. Dashboard values are snapshots from the demonstrated run and should not be interpreted as fixed production benchmarks.

Analytical area	Demonstrated result
Usage	2,072 current usage transactions, 24,818 voice minutes, 9,570 SMS and 3,674.7 GB data in the presentation snapshot.
Incidents	378 network incidents in the illustrated dashboard snapshot.
Network events	898 events available to the CRM dashboard.
Ticket SLA	31.9% compliance against a configured 95% target in the demonstrated filter state.
Packet loss	2.0% average against a 5% dashboard limit.
Critical alarms	89 in the main dashboard snapshot; CRM home separately shows 94 critical events under its own view/filter context.

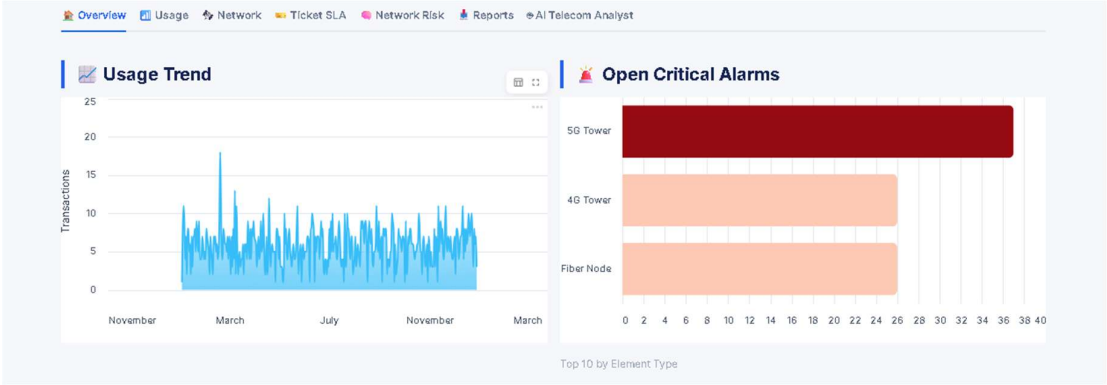


Fig. 13.1: Streamlit overview dashboard with telecom usage and operations KPIs



Fig. 13.2: Incident and packet-loss reliability analysis

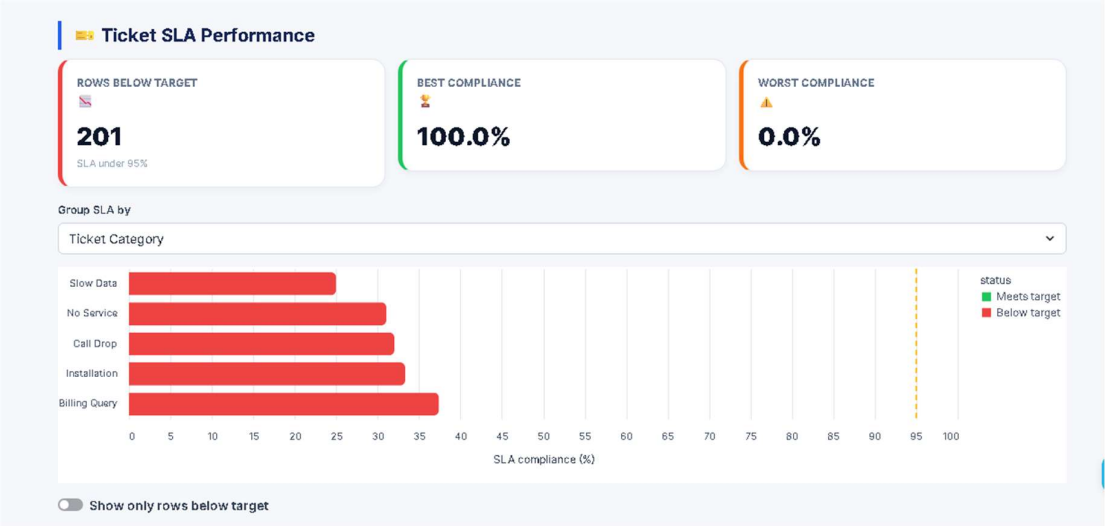


Fig. 13.3: Ticket SLA performance and rows below target

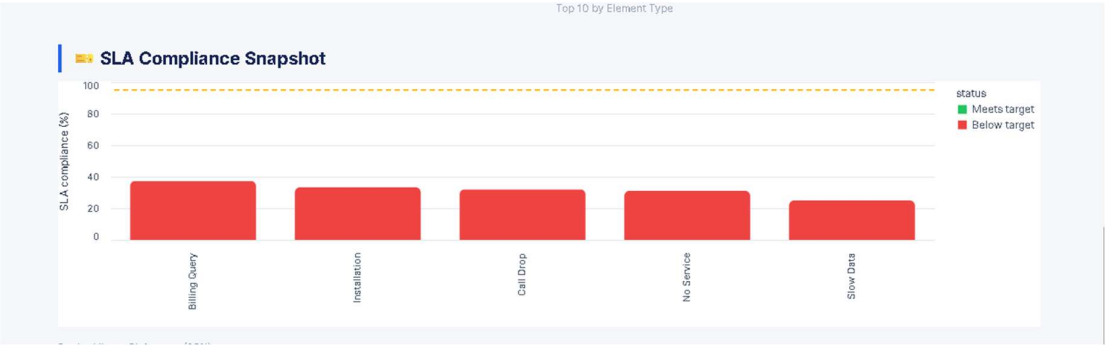


Fig. 13.4: SLA compliance snapshot and category analysis

14. Network Risk Analytics

Network risk is calculated per network element from operational evidence. The canonical implemented design documented in the Gold model uses incident frequency, severity, restore time, open critical alarms and customer-experience impact. A separate dashboard screenshot confirms 361 scored elements, 11 medium-risk elements, no high-risk elements and an average score of 18.8 under the 40/70 presentation bands.

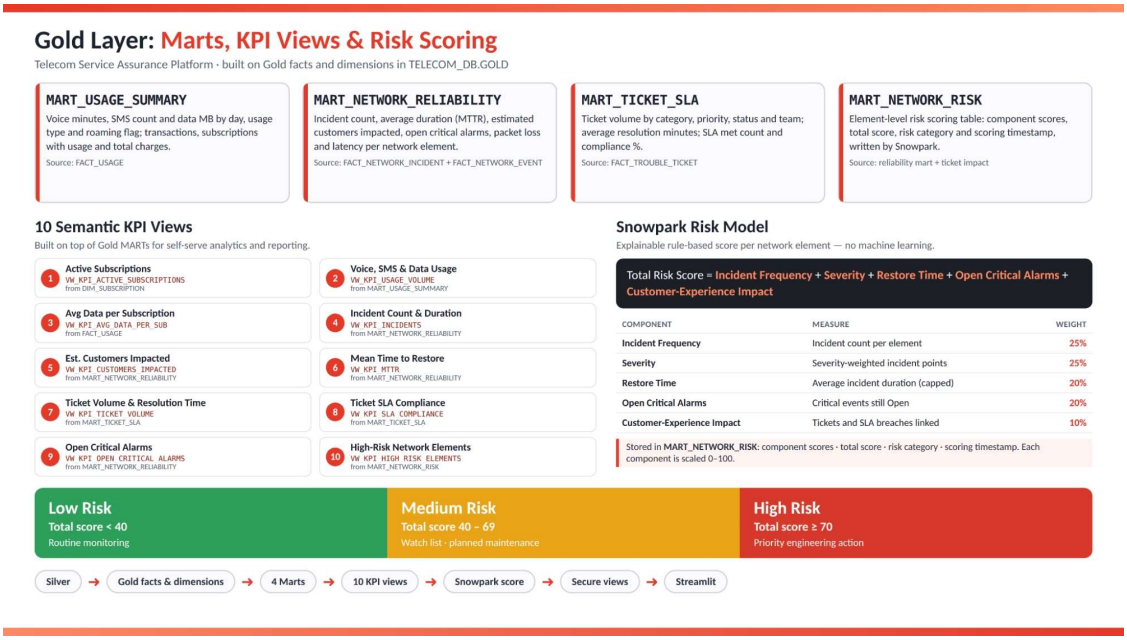


Fig. 14.1: Gold marts, KPI views and explainable network-risk model

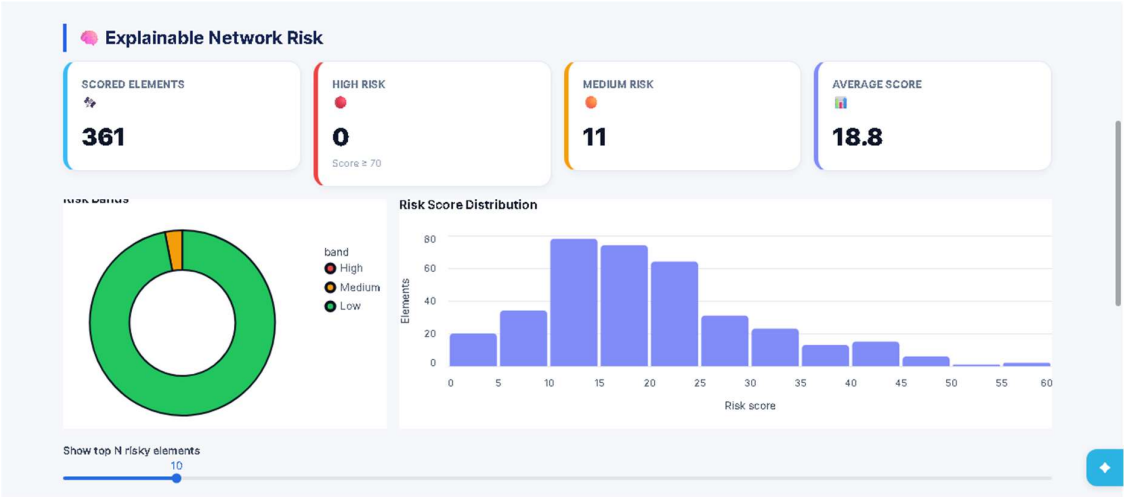


Fig. 14.2: Explainable network-risk distribution and score-band summary

15. KPI Framework

The KPI framework connects business definitions to governed Gold objects. Ten semantic KPI views cover subscriptions, usage, incidents, customer impact, MTTR, tickets, SLA, alarms and high-risk elements.

KPI	Business meaning	Primary source
Active subscriptions	Current active service base	DIM_SUBSCRIPTION
Usage volume	Voice, SMS and data consumption	MART_USAGE_SUMMARY
Average data per subscription	Normalised data usage	FACT_USAGE
Incident count and duration	Operational disruption	MART_NETWORK_RELIABILITY
Estimated customers impacted	Potential service impact	MART_NETWORK_RELIABILITY
MTTR	Average time to restore service	MART_NETWORK_RELIABILITY
Ticket resolution	Volume and time to resolve	MART_TICKET_SLA
SLA compliance	Share of tickets meeting target	MART_TICKET_SLA

Open critical alarms	Urgent unclosed events	MART_NETWORK_RELIABILITY
High-risk elements	Elements exceeding risk threshold	MART_NETWORK_RISK

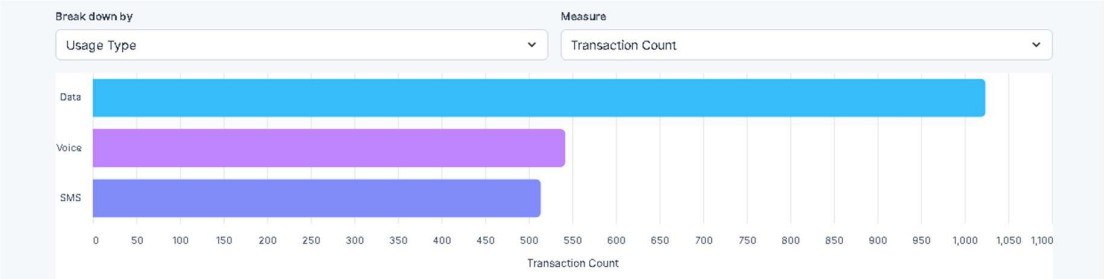


Fig. 15.1: Usage metric breakdown and transaction analysis



Fig. 15.2: Network reliability analysis by element grouping

16. Cortex Analyst Semantic Layer – Project USP

Six semantic views were implemented in TELECOM_DB.SEMANTIC: customer experience, incidents, network events, network reliability, network risk and usage. Cortex Analyst translates a natural-language question into SQL against these governed semantic definitions, and the AI Telecom Analyst tab returns live Snowflake results.

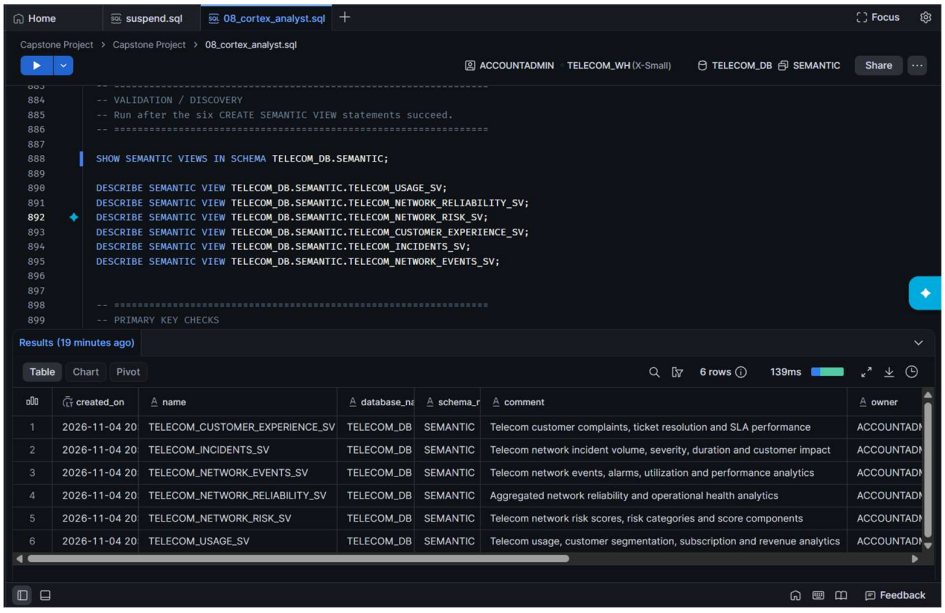


Fig. 16.1: Six semantic views created and validated in the SEMANTIC schema

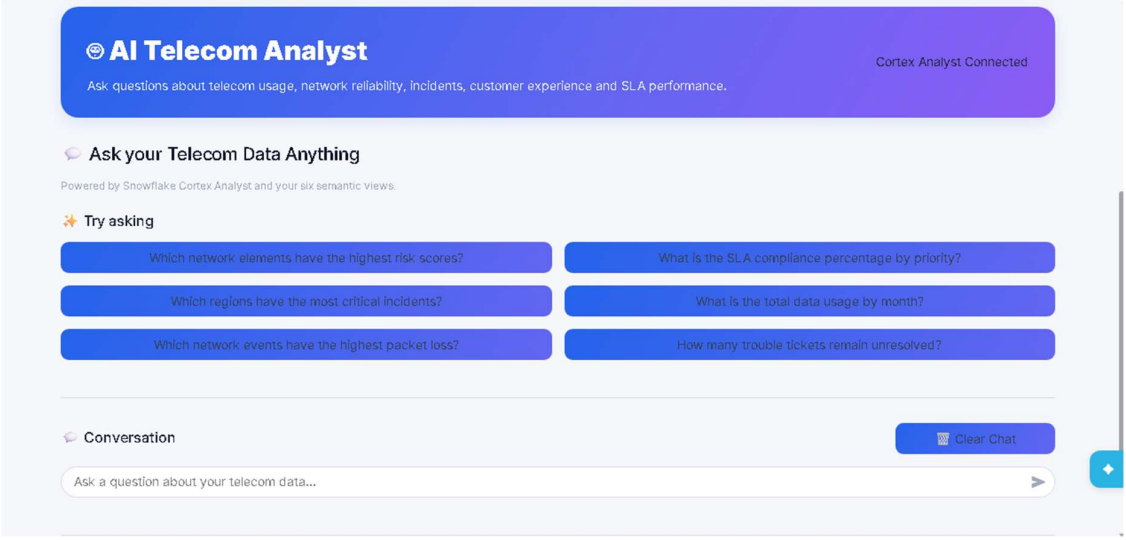


Fig. 16.2: AI Telecom Analyst start screen with suggested telecom questions

17. Automated Critical Incident Alerting – Project USP

The platform does not wait for someone to open a dashboard: it detects unattended critical events and emails the on-call team with full context. The alert runs natively in Snowflake and turns the analytics layer into an operational action within one alert cycle.

Stage	What happens	Detail
1. Detect	Snowflake Alert checks NETWORK_EVENTS every 30 minutes	SEVERITY = CRITICAL, EVENT_STATUS = OPEN, ACKNOWLEDGED = 0
2. Enrich	Joins each event to incidents and trouble tickets	Incident IDs, root cause, customers impacted, past tickets and customers
3. Compose	SEND_TELECOM_CRITICAL_EMAIL builds one block per event	Element, alarm, location, ticket history and recommended action
4. Notify	SYSTEM$SEND_EMAIL via TELECOM_EMAIL_INT	Delivered only to verified, allow-listed recipients

Why it stands out:

- **Proactive:** turns analytics into action within one alert cycle.
- **Context-rich:** engineers see ticket and customer history up front.
- **No noise:** no critical events means no email is sent.
- **Native to Snowflake:** no external scheduler or mail server is needed.

Each email carries one block per critical event, listing the event ID, network element, severity, alarm code, location, historical ticket count, historical ticket and customer IDs, and the action "Investigate immediately."

Component	Purpose
TELECOM_EMAIL_INT	Email notification integration with an allow-listed recipient
Verified recipient user	USERADMIN creates the user; email verification is started
SEND_TELECOM_CRITICAL_EMAIL()	SQL procedure (EXECUTE AS OWNER) builds and sends the report
TELECOM_CRITICAL_ALERT	Runs on COMPUTE_WH every 30 minutes when critical events exist

The alerting was tested in three steps: a SYSTEM$SEND_EMAIL test message confirmed delivery; a query listed the critical, open, unacknowledged events; and the alert was run on demand with ALTER ALERT … RESUME followed by EXECUTE ALERT.

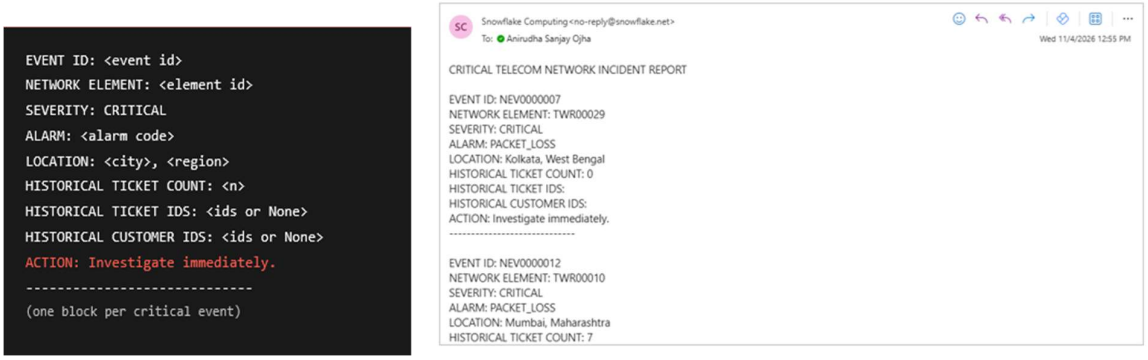


Fig. 17.2: Sample alert email and critical alert emails delivered from Snowflake

The delivered output confirmed the design. All three emails (test message, scheduled alert notice and enriched incident report) arrived from Snowflake on 4 November 2026. The report lists each critical event with its element, alarm and location, and the ticket history adds context, for example seven past tickets for TWR00010 in Mumbai. The team is informed without any manual checking.

18. Snowpark Network-Risk Scoring – Project USP

Snowpark performs transparent rule-based scoring inside Snowflake. The implementation is not a trained machine-learning model. Each component is calculated from domain measures, capped or scaled, combined into a total score, assigned a risk category and written with a scoring timestamp.

Component	Implemented rationale	Model weight in Gold design
Incident frequency	Repeated incidents indicate unstable elements	25%
Severity	Higher-severity incidents receive more weight	25%
Restore time	Slow restoration increases operational exposure	20%
Open critical alarms	Unresolved critical events require immediate attention	20%
Customer-experience impact	Tickets and SLA breaches connect network health to customers	10%

Threshold note: the final evidence includes two historical configurations. The earlier Snowpark slide shows Low below 35, Medium 35 to below 65 and High at 65 or above. The later Gold model and final dashboard use Low below 40, Medium 40–69 and High at 70 or above. This report treats 40/70 as the final presentation configuration and records 35/65 as an earlier implementation variant.

19. Streamlit Dashboard

The primary Streamlit application reads persisted calculations through secure views. SQL and Snowpark own transformations; Streamlit is responsible for filters, KPI cards, charts, tables and drill-down presentation. A second CRM dashboard supports operational alert triage.

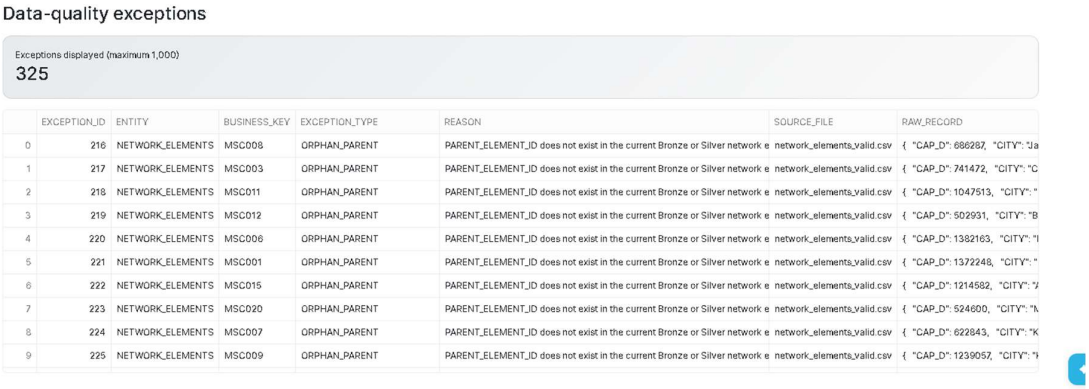


Fig. 19.1: Data-quality exception register displayed in Streamlit

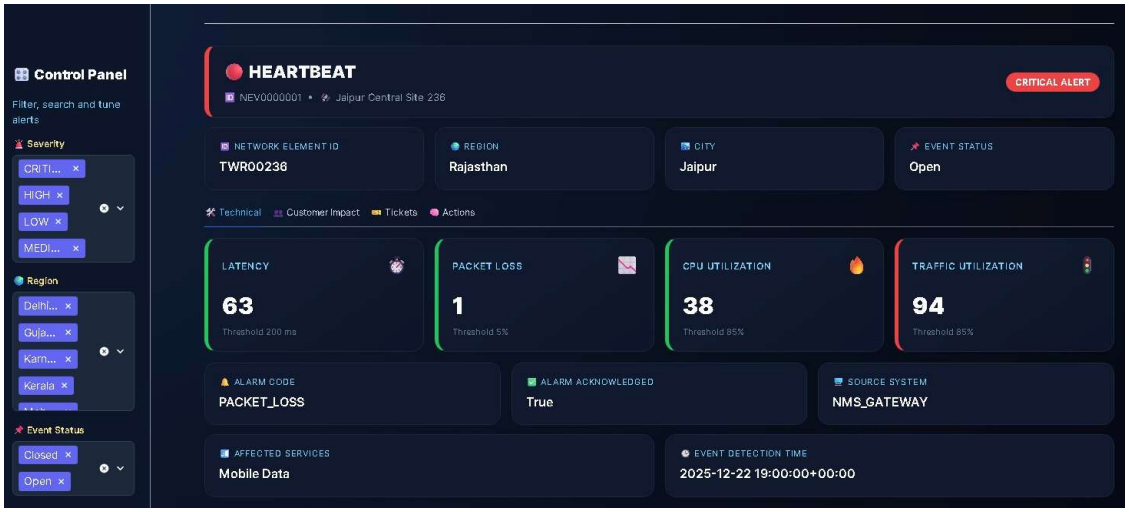


Fig. 19.2: Critical alert investigator with technical thresholds

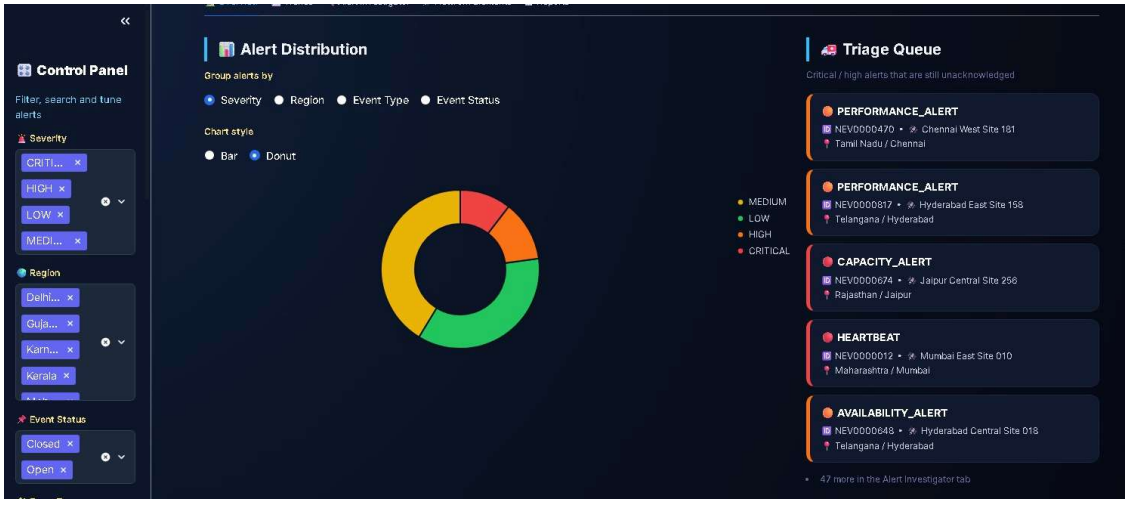


Fig. 19.3: Alert distribution donut and operational triage queue



Fig. 19.4: Alert distribution bar chart and triage queue

20. Technical Flow End-to-End

The runtime sequence begins with delivery of source files to S3, continues through IDMC cleansing and routing, then loads clean files to Snowflake Bronze. Silver enforces typed structures and relationships, Gold applies dimensional logic, and Consumption presents secure outputs. The incremental path separately processes a 10-row Usage file through a Stream and stored procedure before refreshing Gold.

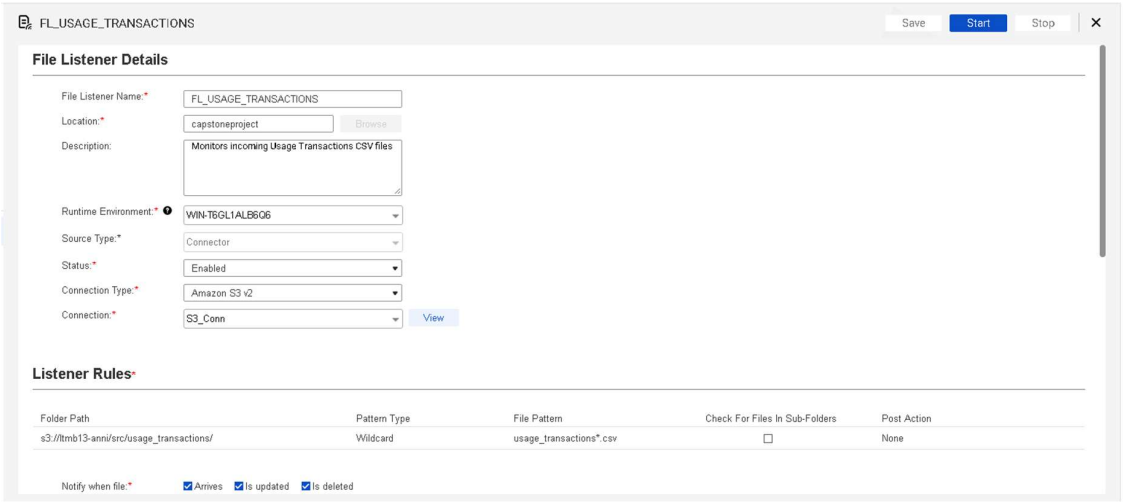


Fig. 20.1: IDMC file listener for incremental usage transactions

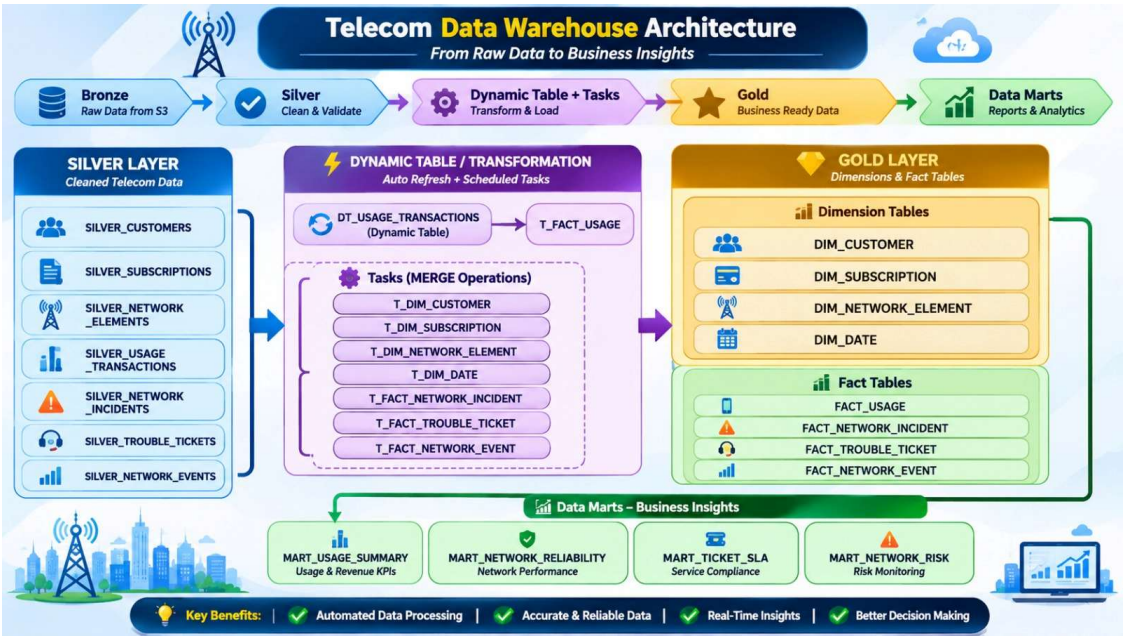


Fig. 20.2: Task and stored-procedure execution for incremental Usage processing

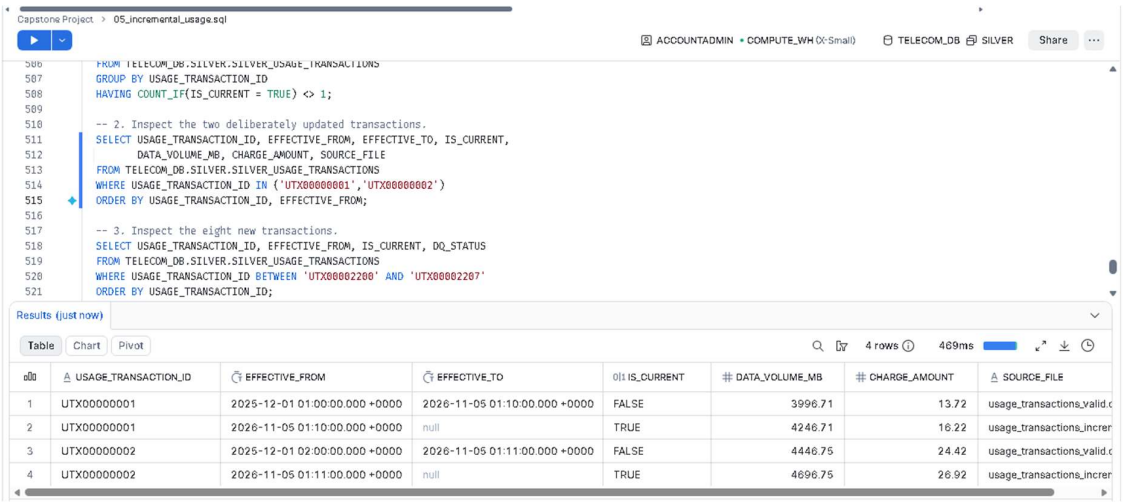


Fig. 20.3: Snowflake warehouse transformation and mart flow

21. Data Quality & Validation

Validation covers load status, source-to-layer counts, controlled data-quality signatures, duplicate keys, referential integrity, SCD Type 2 uniqueness, KPI plausibility, risk-score ranges, dashboard reconciliation, role tests and recovery. The screenshot evidence reports 325 Silver exceptions in the displayed view, largely orphan-parent checks for network elements.

Validation	Expected control
Initial reconciliation	6,900 source rows; 6,886 accepted; 14 excluded upstream.
SCD Type 2	Exactly one IS_CURRENT = TRUE row per usage_transaction_id.
Risk scores	Populated, explainable and bounded.
Referential integrity	No orphan Gold fact-to-dimension surrogate keys.
Replay	Unchanged replay does not add a duplicate current version.
Dashboard	Displayed totals reconcile to the underlying Gold or Consumption object for the same filter state.

Data-quality exceptions

Exceptions displayed (maximum 1,000)
325

	EXCEPTION_ID	ENTITY	BUSINESS_KEY	EXCEPTION_TYPE	REASON	SOURCE_FILE	RAW_RECORD
0	216	NETWORK_ELEMENTS	MSC008	ORPHAN_PARENT	PARENT_ELEMENT_ID does not exist in the current Bronze or Silver network e	network_elements_valid.csv	{ "CAP_D": 686287, "CITY": "Ja
1	217	NETWORK_ELEMENTS	MSC003	ORPHAN_PARENT	PARENT_ELEMENT_ID does not exist in the current Bronze or Silver network e	network_elements_valid.csv	{ "CAP_D": 741472, "CITY": "C
2	218	NETWORK_ELEMENTS	MSC011	ORPHAN_PARENT	PARENT_ELEMENT_ID does not exist in the current Bronze or Silver network e	network_elements_valid.csv	{ "CAP_D": 1047513, "CITY": "
3	219	NETWORK_ELEMENTS	MSC012	ORPHAN_PARENT	PARENT_ELEMENT_ID does not exist in the current Bronze or Silver network e	network_elements_valid.csv	{ "CAP_D": 502931, "CITY": "B
4	220	NETWORK_ELEMENTS	MSC006	ORPHAN_PARENT	PARENT_ELEMENT_ID does not exist in the current Bronze or Silver network e	network_elements_valid.csv	{ "CAP_D": 1382163, "CITY": "
5	221	NETWORK_ELEMENTS	MSC001	ORPHAN_PARENT	PARENT_ELEMENT_ID does not exist in the current Bronze or Silver network e	network_elements_valid.csv	{ "CAP_D": 1372248, "CITY": "
6	222	NETWORK_ELEMENTS	MSC015	ORPHAN_PARENT	PARENT_ELEMENT_ID does not exist in the current Bronze or Silver network e	network_elements_valid.csv	{ "CAP_D": 1214582, "CITY": "
7	223	NETWORK_ELEMENTS	MSC020	ORPHAN_PARENT	PARENT_ELEMENT_ID does not exist in the current Bronze or Silver network e	network_elements_valid.csv	{ "CAP_D": 524600, "CITY": "M
8	224	NETWORK_ELEMENTS	MSC007	ORPHAN_PARENT	PARENT_ELEMENT_ID does not exist in the current Bronze or Silver network e	network_elements_valid.csv	{ "CAP_D": 622843, "CITY": "K
9	225	NETWORK_ELEMENTS	MSC009	ORPHAN_PARENT	PARENT_ELEMENT_ID does not exist in the current Bronze or Silver network e	network_elements_valid.csv	{ "CAP_D": 1239057, "CITY": "

Fig. 21.1: Silver data-quality exceptions with reason and source-file traceability

Instance Name	Location	Subtasks	Start Time	End Time	Rows Processed	Status
m_network_in...	capstoneproject		Nov 2, 2026, 7:56 ...	Nov 2, 2026, 7:56 ...	400	Success
M_network_ev...	capstoneproject		Nov 2, 2026, 5:13 ...	Nov 2, 2026, 5:14 ...	900	Success
M_customers-1	capstoneproject		Nov 2, 2026, 5:00 ...	Nov 2, 2026, 5:00 ...	899	Success
M_network_el...	capstoneproject		Nov 2, 2026, 4:55 ...	Nov 2, 2026, 4:55 ...	400	Success
usage_transa...	capstoneproject		Nov 2, 2026, 4:44 ...	Nov 2, 2026, 4:44 ...	2199	Success
M_Subscriptio...	capstoneproject		Nov 2, 2026, 4:36 ...	Nov 2, 2026, 4:37 ...	1400	Success
M_Trouble_Tic...	capstoneproject		Nov 2, 2026, 4:26 ...	Nov 2, 2026, 4:27 ...	700	Success

Fig. 21.2: Successful project execution and processed-row validation

22. Security & Governance

The security model applies least privilege through role separation, masking policies and secure views. NETWORK_ENGINEER_ROLE focuses on operational engineering access; SERVICE_ASSURANCE_ROLE consumes masked analytical outputs; TELECOM_MANAGER_ROLE receives curated visibility. Customer email, phone and address are protected by role-aware masking.

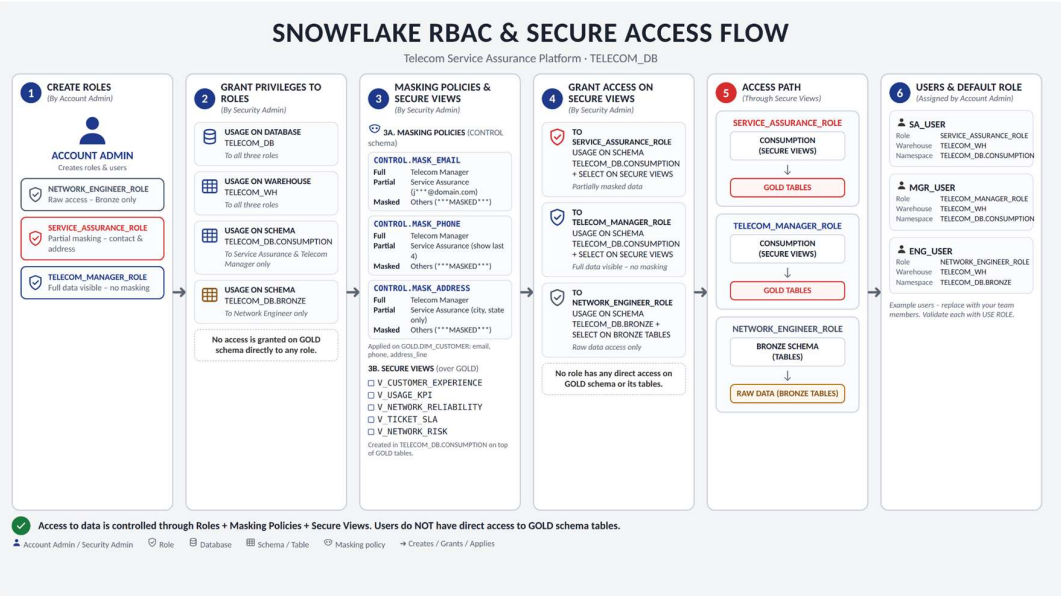


Fig. 22.1: Snowflake RBAC, masking policies, secure views and user access flow

Governance controls include source metadata, exception logging, object ownership, secure reporting boundaries, Time Travel or CLONE recovery, and explicit USE ROLE tests. A future production hardening step is to replace broad Consumption grants with object-specific grants required by each application.

23. Performance & Scalability

The project uses design choices that scale beyond the capstone volume: append-only Streams isolate changed rows, Tasks run only when a Stream contains data, MERGE avoids rebuilds, views propagate refreshed facts to marts, and calculations remain inside Snowflake. Warehouse sizing in screenshots is X-Small, suitable for the demonstrated volume but not a capacity recommendation for production.

- Use query profiles before changing warehouse size or clustering.
- Retain selective projections in secure views to reduce scanned columns.
- Schedule Tasks with SYSTEM$STREAM_HAS_DATA to prevent empty runs.
- Partition operational responsibility: Snowpark calculates risk; Streamlit only reads persisted results.
- For production, retain score history in an append-only fact rather than overwriting the latest snapshot.

24. Error Handling & Edge Cases

The stored-procedure design protects good rows from bad-row failures. TRY_TO_* functions convert parsing failures to NULL for routing. Missing keys, bad timestamps and orphan relationships are written to the exception register. Transactions ensure that COMMIT advances a successful batch, while ROLLBACK preserves the Stream offset for retry.

Edge case	Response
Same unchanged record replayed	No additional SCD version.
Existing ID with changed payload	Close old version and insert new current version.
Older incoming version	Log as stale and leave current row unchanged.
Missing parent network element	Write an ORPHAN_PARENT exception.
No qualifying critical events	Do not send an email.
Task created before manual validation	Keep suspended until the procedure has been exercised successfully.

25. Challenges & Solutions

The implementation addressed both technical and delivery constraints within a five-day sprint.

Challenge	Solution	Residual consideration
Seven heterogeneous sources	IDMC mappings with dataset-specific rules and common routing	Automate regression profiling when source contracts change.
Historical Usage corrections	Stream plus row hash and SCD Type 2 procedure	Keep one current row and validate replay continuously.
Conflicting risk thresholds	Document final 40/70 bands and preserve earlier 35/65 as history	Centralise thresholds in a configuration table.
Sensitive customer fields	Masking policies and secure views	Tighten grants to exact objects.
Operational alert overload	Filter to critical, open, unacknowledged events and enrich context	Add suppression and escalation windows.

Dashboard count differences	Treat screenshots as filter-state snapshots and reconcile to source views	Display “as of” timestamp and active filters prominently.

26. Sprint Execution & Project Management

The project was managed as a five-day sprint using a Kanban board and a burndown chart. Work moved from profiling and IDMC ingestion through Snowflake layers, incrementals, analytics, security and final demonstration. Task ownership was visible on the board, while the burndown tracked remaining work against an ideal trajectory.

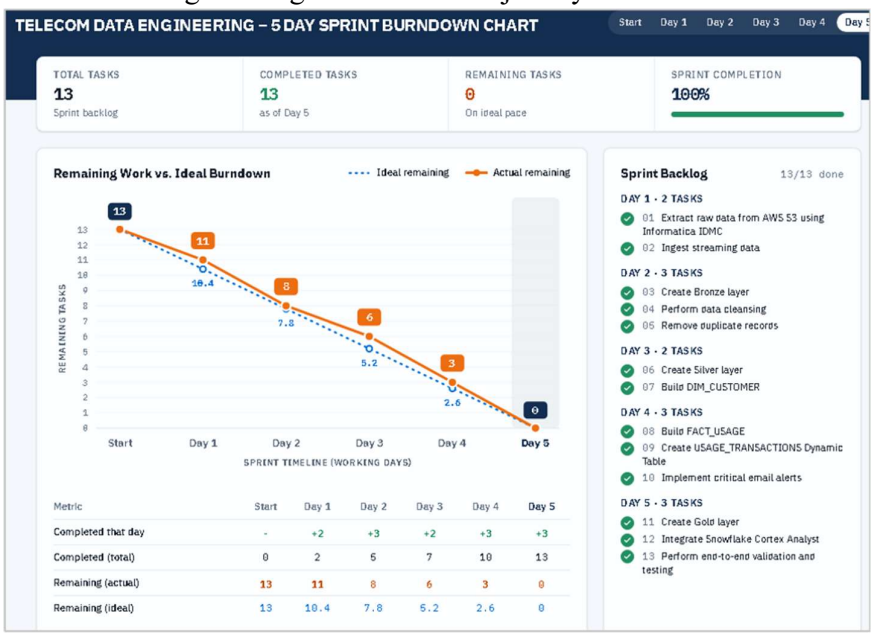


Fig. 26.1: Five-day sprint burndown with 13 of 13 tasks completed

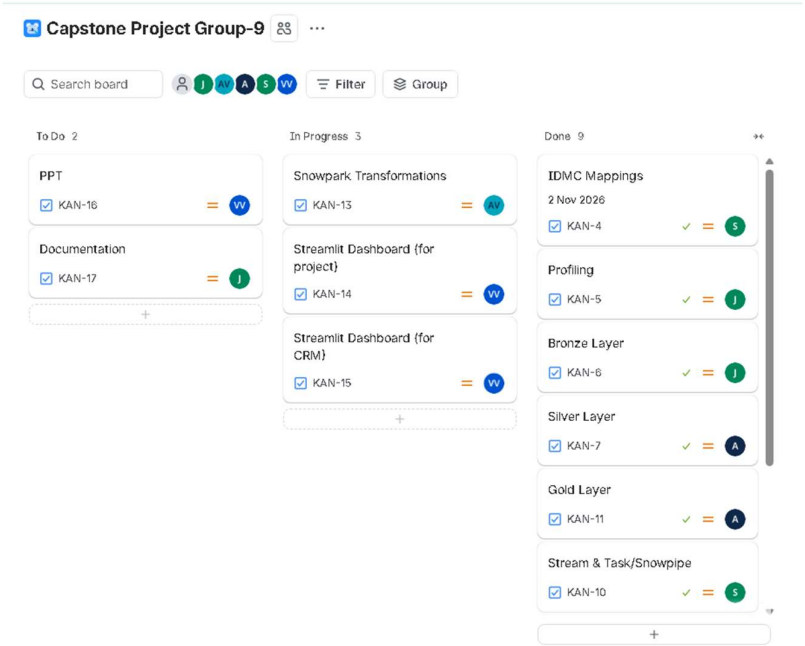


Fig. 26.2: Capstone Project Group-9 Kanban board and task status

Day	Execution focus	Progress evidence
Day 1	Domain discovery, architecture, source profiling and mapping design	2 tasks completed; 11 remaining.
Day 2	Bronze creation, cleansing and duplicate handling	3 tasks completed; 8 remaining.
Day 3	Silver layer and initial dimensional work	2 tasks completed; 6 remaining.
Day 4	Gold facts, incremental processing and critical alerting	3 tasks completed; 3 remaining.
Day 5	Gold completion, Cortex Analyst, validation and testing	3 tasks completed; 0 remaining; 100%.

27. Deliverables Achieved

The delivered solution exceeds the original baseline in several areas while staying within the telecom domain.

- Seven-source profiling and IDMC cleansing/routing evidence.
- Snowflake Bronze, Silver, Gold, Control, Consumption and Semantic implementation.
- Four dimensions, four facts, four core analytical marts and ten KPI views.
- Incremental Usage pipeline with Stream, Task, stored procedure, SCD Type 2 and Gold MERGE.
- Explainable Snowpark network-risk scoring.
- Main Streamlit analytics application and dedicated Telecom CRM alert dashboard.
- Six Cortex Analyst semantic views and embedded AI Telecom Analyst experience.
- RBAC, masking policies, secure views, exception handling and recovery approach.
- Native critical-event email alert workflow and execution evidence.

28. Overall Project Assessment

The project is technically coherent and demonstrable: data-quality responsibilities are separated, the warehouse model has explicit grains, incremental processing preserves history safely, risk is explainable, and consumption is governed. The strongest differentiators are the Cortex Analyst layer, native critical-event alerting and the operational CRM dashboard.

Strengths

- Clear separation between initial batch ingestion and incremental Usage processing.
- Strong traceability from source files through exceptions and Gold outputs.
- Rule-based risk is transparent and appropriate for a capstone without labelled training data.
- Operational alerting converts analytics into action rather than relying only on dashboards.

Areas for Future Iteration

- Centralise thresholds and KPI targets in governed configuration tables.
- Retain historical risk snapshots rather than replacing the latest score table.
- Automate DIM_DATE extension and production deployment pipelines.
- Tighten object grants and complete repeatable role-test automation.

Future Enhancements

- Continuous event ingestion using Snowpipe Streaming or an approved event platform.
- Alert suppression, deduplication and escalation policies.
- Historical trend analysis for risk movement and SLA recovery.
- Production monitoring for Task failures, Stream lag, query cost and dashboard freshness.
- published test results were fabricated.